Exploratory Analyses I

Today's agenda:

Get the course repository

Discuss data of interest

Exploratory analyses and graphics

Work through 2.6

Exploratory Analyses and Graphics

Why do we do exploratory analyses and graphing?

Exploratory Analyses and Graphics

Why do we do exploratory analyses and graphing?

- Detect errors
    - E.g. nonsensical values, perfect correlations, “weird” patterns
- Understand statistical distributions/ check for violation of assumptions
    - E.g., non-normal distributions where normal are assumed
- Find patterns and relationships
    - E.g., mapping variables, pairwise correlation plots, etc.
- Identify variables to drop
    - E.g., highly correlated variables (in some cases)

Exploratory Analyses and Graphics

Critical part of science!

Getting data from Github

Last time you downloaded one file

Today you need several, and I'll add more all semester.

There's a better way: **get the whole repo at once, and update it with one click.**

github.com/bmaitner/biometry_course

Cloning a repo

Same as your own repo in Lecture 2, different URL

In RStudio:

File → New Project… → Version Control → Git

Paste https://github.com/bmaitner/biometry_course

Pick a folder. Create Project.

(Big download, we'll continue while it runs)

Why a Project, not just a folder

Class scripts use paths like:

```
read.csv("data/Avonet/AVONET1_BirdLife.csv")
```

Those only resolve if R is *in* the correct directory

Opening it as a Project is what puts you there.

None of the confusion of looking for your files!

Getting updates later

I add lectures, scripts and data as we go.

To get those updates:

Git pane → Pull (the blue down arrow)

Do this before every class.

Don't edit existing files in this folder

If you change a script here, the next Pull may refuse to run.

(this can be fixed, but it’s a hassle)

Instead: copy what you want into your project, then edit.

Alternatively, you can create a folder with your name on it

- Nothing in there should be overwritten

Treat the code and data I’ve provided as read-only.

What types of data are we interested in?

Let's go around and discuss types of data and questions of interest.

(Reminder: you'll be selecting one or more focal datasets to examine)

In particular, let us know:

- If you already have data
- If you need help finding data

Note: Have a look in `data/` in the repo you just cloned, each folder has a README.

Exploratory Graphics

Which plot?

It depends on what kind of variables you have.

Predictor	Response	Plot
none	One continuous	`hist, boxplot`
One categorial	One continuous	`boxplot`
One continuous	One continuous	`plot`
none	One categorical	`table, barplot`
Two categorical	-	`table, mosaicplot`

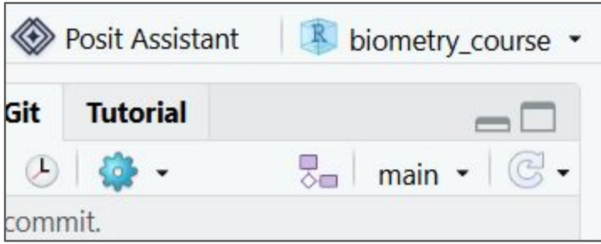
(Bolker's Table 2.1 is the long version)

Start with one continuous variable

First, load the avonet data in as an object called “avonet”

Does everyone remember how to do that?

Make sure you're in the biometry_course project: check the top right



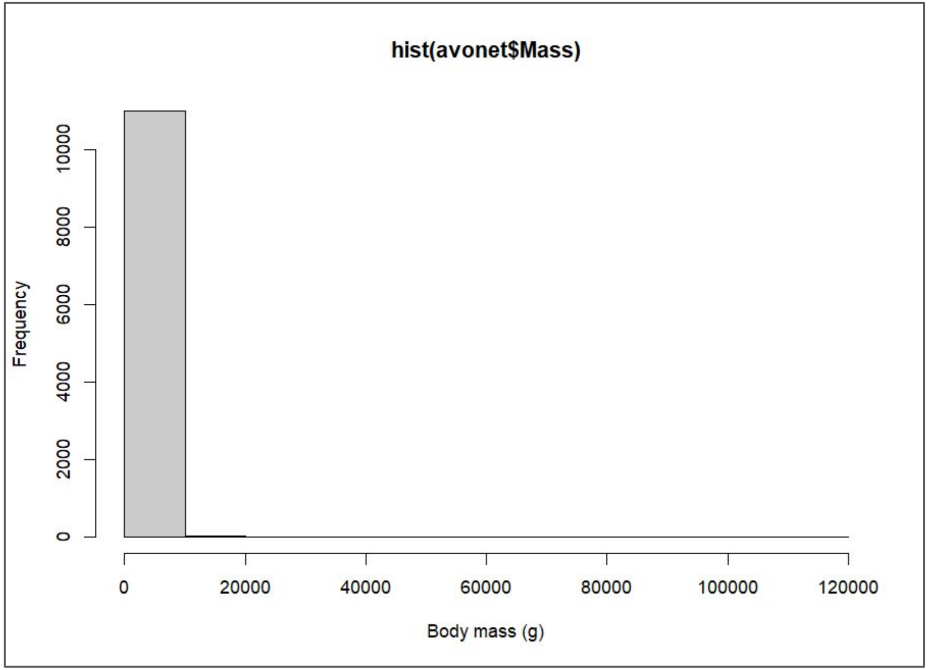
Then, view a histogram of mass values using:

hist(avonet$Mass)

That's one bar

This is not broken. It's telling you the truth, badly.

10,995 of 11,009 species (99.9%) are in that first bin



Why

Try:

```
range(avonet$Mass)
```

```
[1]     1.9 111000.0
```

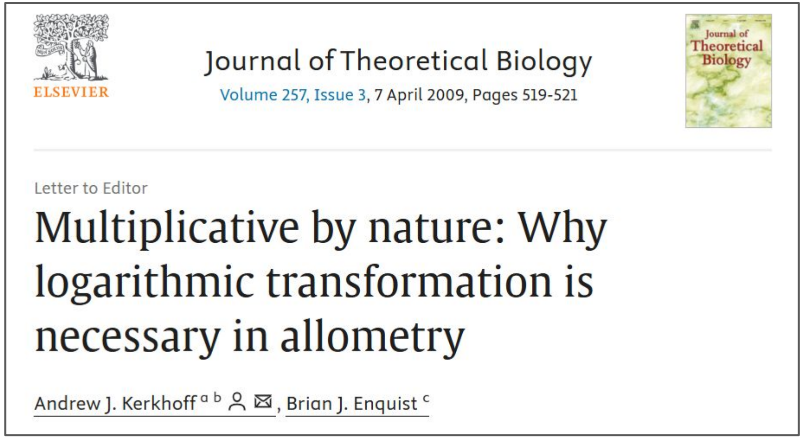
Peruvian sheartail: 1.9 g Ostrich: 111,000 g

A 58,000-fold range. Median mass is 35.5 g.

Log scaling and body size

Run:

```
hist(log10(avonet$Mass),
    breaks = 40)
```



ELSEVIER

Journal of Theoretical Biology

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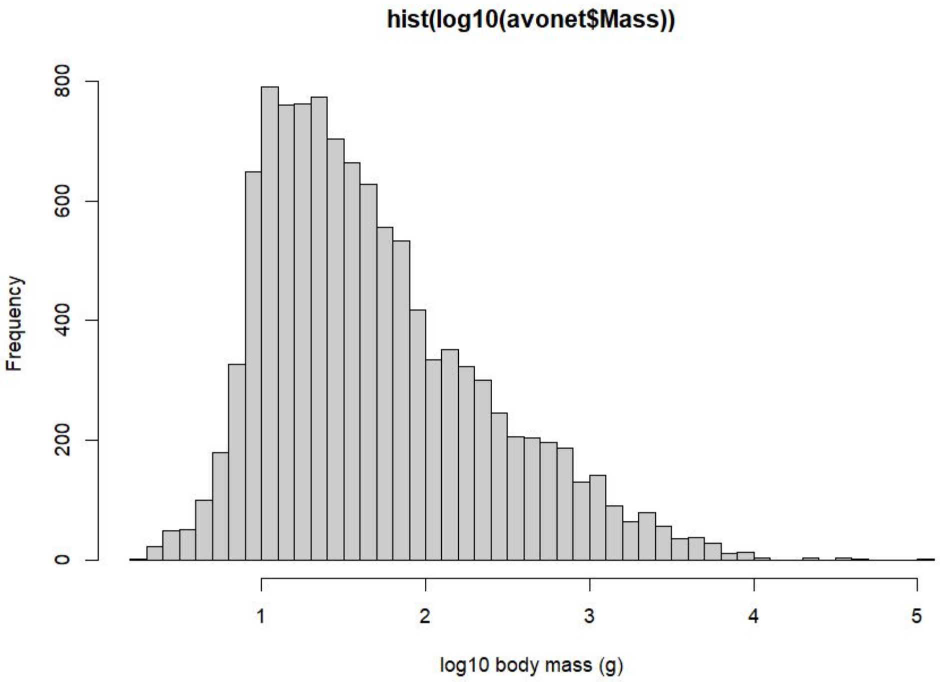
Letter to Editor

Multiplicative by nature: Why logarithmic transformation is necessary in allometry

Andrew J. Kerkhoff [a b], Brian J. Enquist [c]

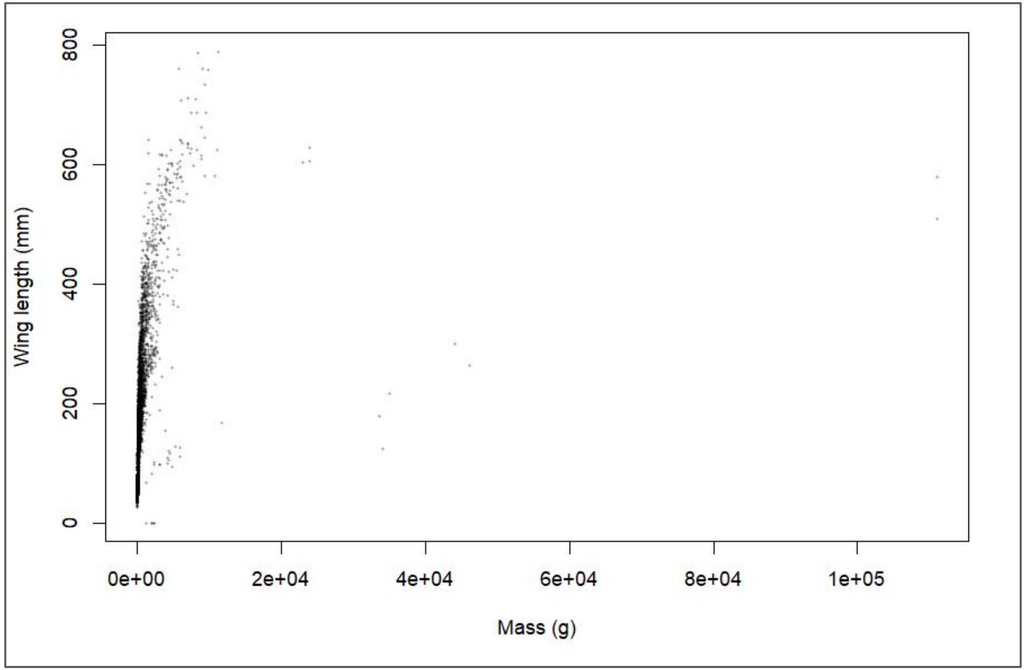


Two continuous variables

Now:

```
plot(x = avonet$Mass,
     y = avonet$Wing.Length)
cor(avonet$Mass,
    avonet$Wing.Length,
    use = "complete.obs")
```

What is the correlation?



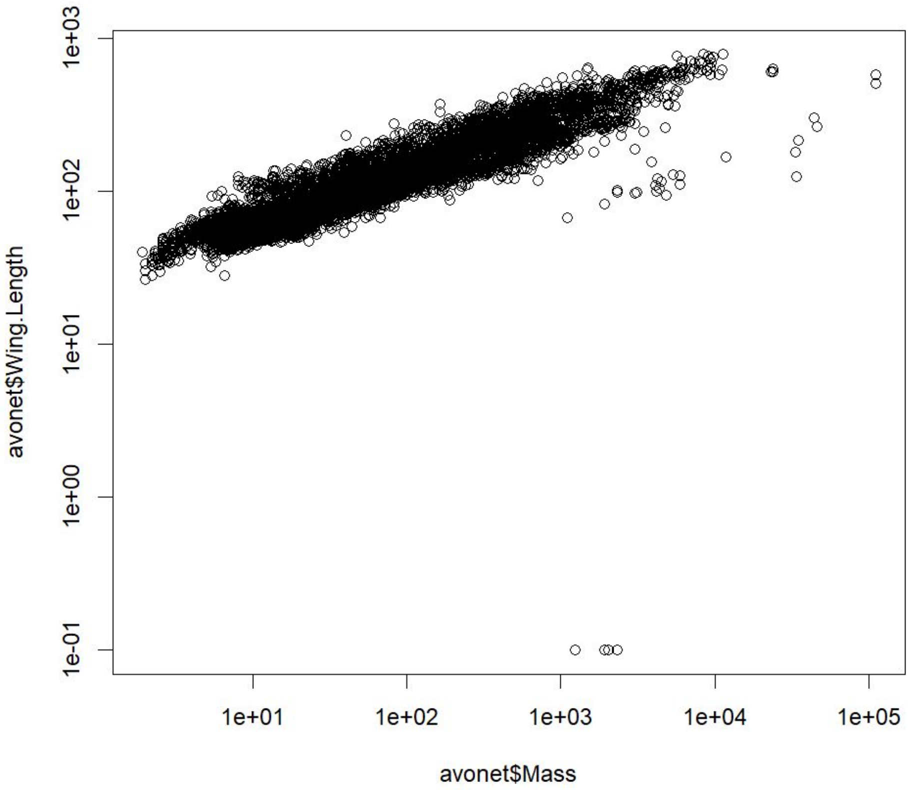
The same data, log-log

Try using a log-log plot:

```
plot(x = avonet$Mass,
     y = avonet$Wing.Length,
     log = "xy")
```

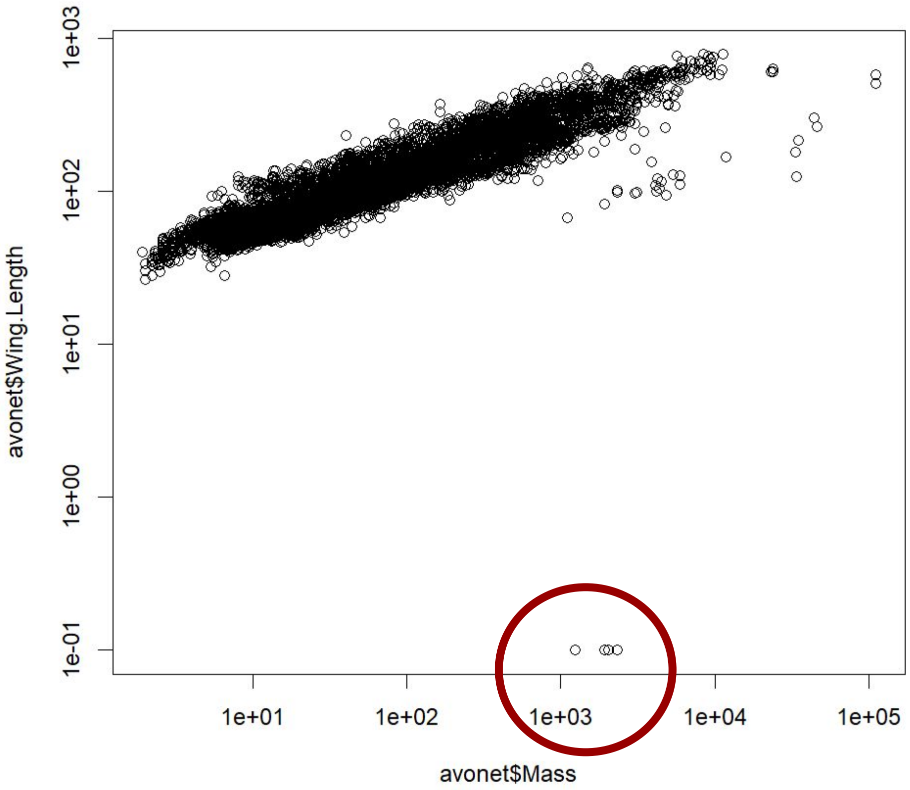
What is the correlation now?

Hint: use the `log()` function



What birds are those?

Any guesses?



Kiwis

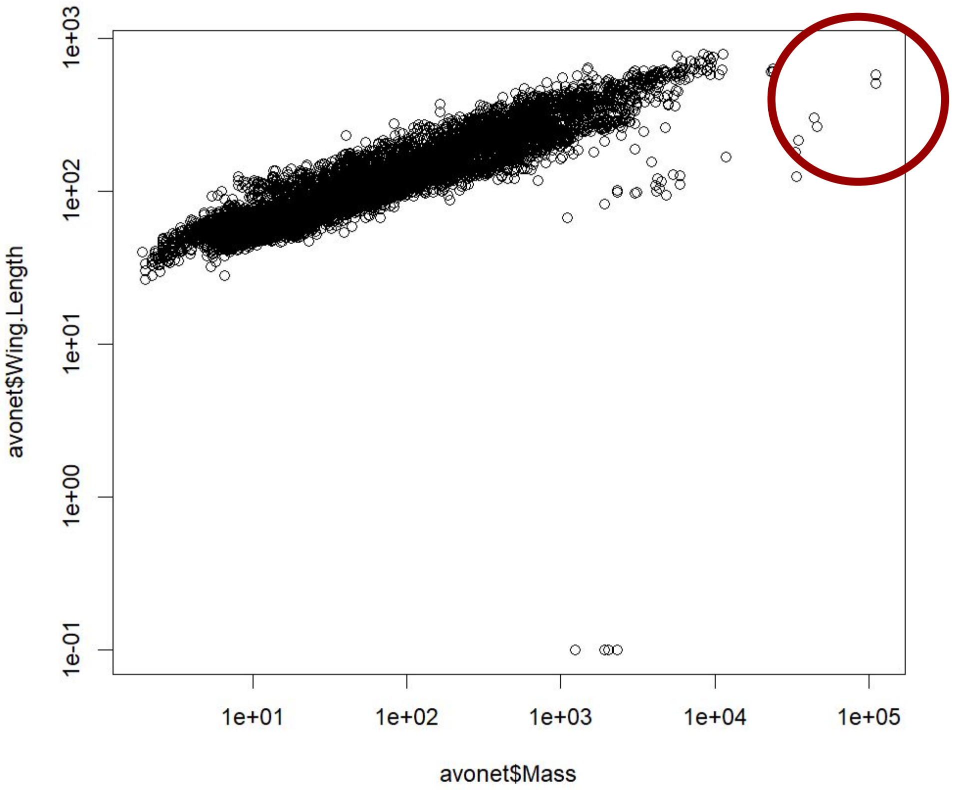
```
avonet[which(avonet$Wing.Length < 1),
       c("Species1", "Family1",
         "Mass", "Wing.Length")]
```

```
                Species1     Family1   Mass Wing.Length
10853 Apteryx australis Apterygidae 2320.5         0.1
10854   Apteryx haastii Apterygidae 2022.7         0.1
10855  Apteryx mantelli Apterygidae 2320.5         0.1
10856    Apteryx owenii Apterygidae 1238.3         0.1
10857      Apteryx rowi Apterygidae 1924.0         0.1
```

0.1 is a placeholder for "too small to measure."

What birds are those?

Any guesses?



Outlier ≠ error

Try this:

```
avonet[which(avonet$Mass > 20000),
       c("Species1", "Mass", "Wing.Length")]
```

```
                       Species1     Mass Wing.Length
10597        Aptenodytes forsteri  33569.3       179.4
10858          Casuarius bennetti  35000.0       216.8
10859         Casuarius casuarius  44000.0       300.8
10860 Casuarius unappendiculatus  46073.9       264.5
10861    Dromaius novaehollandiae  34093.3       124.5
10862              Rhea americana  23000.0       604.5
10863                Rhea pennata  23900.0       629.5
10864           Rhea tarapacensis  23900.0       607.0
10865            Struthio camelus 111000.0       579.0
10866       Struthio molybdophanes 111000.0       510.5
```

Also extreme. Also flightless. These are real measurements.

Too many points

```
plot(x = avonet$Mass,
     y = avonet$Wing.Length,
     log = "xy",
     pch = 16,
     cex = 0.4,
     col = rgb(0, 0, 0, alpha = 0.3))
```

cex: controls point size

alpha: controls transparency

Now you can see where the density is.

A categorical predictor

Try this out:

```
boxplot(log10(Mass) ~ Trophic.Level,
        data = avonet)
```

What trophic level is the largest?

Notes:

~ means "as a function of". You'll see this notation all semester.

Used “data” argument rather than “avonet$” approach

One categorical variable

Try both approaches. Which is more informative?

```
table(avonet$Habitat)
barplot(sort(table(avonet$Habitat),
             decreasing = TRUE),
        las = 2)
```

Note: las = 2 turns the labels sideways (Label Axis Style)

Numbers that aren't quantities

Migration is coded 1, 2, 3 (sedentary / partial / migratory)

Try:

```
mean(avonet$Migration)

mean(avonet$Migration, na.rm = TRUE)
```

What do you get?

Numbers that aren't quantities

R doesn’t know meaning, that’s up to you.

What class is R treating migration as?

Potential solutions (try them out):

Use table:

```
barplot(table(avonet$Migration))
```

`Correct the class:`

```
avonet$Migration <- as.factor(avonet$Migration)
```

Remainder of class:

Work through 2.6. Start at 2.6.2 (the plots), but first run:

```
install.packages("emdbook")   # once
library(emdbook)
data(SeedPred)
str(SeedPred)
```

Skip the reconstruction in 2.6.1 — one line gets you the finished dataset.

Next time:

Before class:

- review 2.5 - 2.6

During class:

- Discuss Assignment 1
- Discuss potential data sources to work with
- Continue working through 2.6 (if needed)
- Start to apply techniques from chapter 2 to dataset of interest